Mehmet Yigit Balik

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Mehmet Yigit Balik | YigitBalik | Yigit Balik

Education

Aalto University — Espoo, Uusimaa, Finland
Ph.D. in Computer Science — 06 2025 – 06 2029
- Thesis Topic: Interpretable and Scalable Modeling of Dynamic Processes with Deep Generative Models

Aalto University — Espoo, Uusimaa, Finland
M.Sc. in Machine Learning, **GPA - 5/5** — 08 2023 – 06 2025
- Granted 100% Scholarship.
- Thesis: *Bayesian Basis Function Approximation for Scalable Gaussian Process Priors in Deep Generative Models.*

Istanbul Technical University — Istanbul, Turkey
B.Sc. in Computer Engineering, **GPA - 3.91/4** — 08 2019 – 06 2023
- Ranked **3rd** at the faculty **4th** at the university.
- Graduation Project: *Interpretable Deep Learning for Myocardial Infarction Detection from ECG Signals.*

Experience

Aalto University — Espoo, Uusimaa, Finland
Researcher — 06 2024 – Present
- Conducting research on **interpretable and scalable modeling of dynamic processes with deep generative models** under **Dr. Harri Lähdesmäki** in the **Computational Systems Biology Research Group** .

Predictive Intelligence & Medical Imaging Laboratory — Istanbul, Turkey
Undergraduate Student Researcher — 02 2022 – 09 2023
- Worked on electrocardiogram signals using **explainable deep learning** models under the supervision of **Dr. İlkay Öksüz**.
- **Collaborated** with four cardiologists to validate our outcomes.
- Developed **end-to-end** deep learning model and application for medical practitioners.
- Achieved **98.66%** F1 score on PTB-XL dataset in the detection of myocardial infarction.
- Actively contributed to weekly **journal club** meetings.

Brain And Signal Research & Analysis Laboratory — Istanbul, Turkey
Undergraduate Student Researcher — 06 2022 – 10 2022
- Investigated the reproducibility of **federated graph neural networks** on medical datasets under the supervision of **Dr. Islem Rekik**.
- Published paper was **shortlisted** for the **best paper award** (PRIME MICCAI 2022).

Delivers AI — Istanbul, Turkey
Software Engineering Intern — 08 2021 – 09 2021
- Contributed to Robot Management Application by using **ReactJS** and **.NET**.

OBSS Technology — Istanbul, Turkey
Software Engineering Intern — 07 2021 – 08 2021
- Developed a **MVC** based full-stack web app using **Java Spring Boot**, **Hibernate**, **React** and **MySQL**.
- My internship project ranked among the **top three** out of 40 internship projects.

Skills

Coding — Python, C, C++, Java, JavaScript, SQL, NoSQL

Skills (continued)

Technologies — Linux, React, Spring, Flask, TensorFlow, PyTorch, ScikitLearn, Git

Research Publications

1. M. Y. **Balık** and H. Lähdesmäki, "Modeling temporal scrna-seq data with latent gaussian process and optimal transport," in *Forty-third International Conference on Machine Learning*, 2026.
2. M. Y. **Balık**, M. Sinelnikov, P. Ong, and H. Lähdesmäki, "Bayesian basis function approximation for scalable gaussian process priors in deep generative models," in *Forty-second International Conference on Machine Learning*, 2025.
3. T. Tanyel, S. Atmaca, K. Gökçe, M. Y. **Balık**, A. Güler, E. Aslanger, and İ. Öksüz, "Interpretable ecg analysis for myocardial infarction detection through counterfactuals," *Biomedical Signal Processing and Control*, vol. 102, p. 107 227, 2025.
4. M. Y. **Balık**, K. Gökçe, S. Atmaca, E. Aslanger, A. Güler, and İ. Öksüz, "Interpretable deep learning for myocardial infarction detection from ecg signals," in *2023 31st Signal Processing and Communications Applications Conference (SIU)*, IEEE, 2023, pp. 1–4.
5. M. Y. **Balık**, A. Rekik, and I. Rekik, "Investigating the predictive reproducibility of federated graph neural networks using medical datasets," in *Predictive Intelligence in Medicine: 5th International Workshop, PRIME 2022, Held in Conjunction with MICCAI 2022, Singapore, September 22, 2022, Proceedings*, Springer, 2022, pp. 160–171.

Volunteer Experience

Slush — Helsinki, Finland
Badge Claim Volunteer — 11 2023 – 11 2023
- Helped attendees to register for the event and prepare their badges.

ITU ACM Student Chapter — Istanbul, Turkey
Administrative Board Member and Educational Instructor — 08 2021 – 04 2023
- Helped the organization of an algorithm competition that received **1053** applications.
- **Instructed** C programming course to nearly **40** students in the online sessions.
- Helped the foundation of the algorithm team and mentored 12 students.
- Prepared algorithm homeworks.

Projects

Deep Generative Basis Function Gaussian Process — 07 2025
- Implemented in **PyTorch**.
- Designed for interpretable and scalable modeling of longitudinal data.

LLM Assisted RFP Processing Pipeline — 05 2024
- Built a **RAG** pipeline using Upstash vector search and OpenAI to automate RFP proposal generation from PDFs.
- Integrated Hugging Face embeddings to enable semantic search and context retrieval within a multi-stage LLM workflow.

Real-Life MI Detection System — 05 2023
- Implementation and application of B.Sc. graduation project.
- Easy to use app for cardiologists to obtain interpretable classification from ECG reports.
- Model implemented in **TensorFlow**.
- Application developed using **Flask** and **JavaScript**.

Projects (continued)

- **EventWise** 12 2022
  - A complete mobile application that allows people to manage event groups.
  - Developed the backend and database operations using **Spring Boot**, **Hibernate** and **AWS** in a **four-person team**.
- **Federated Reproducibility** 09 2022
  - Implemented in **PyTorch Geometric**.
  - Connectomic and biomedical image datasets are used.

Certificates

- **Inzva Algorithm Competition Summer Camp Participation** 06 2022
- **Inzva Algorithm Program '21-22 Spring Participation** 04 2022
- **Inzva Algorithm Competition Winter Camp Participation** 02 2022

References

Available on request